2)

$F_1 = F_x(1200)\ N$

$F_2 = F_x(\cos 40°) + F_y(\text{sen}\ 40°)$

$\rightarrow F_2 = F_x(689,4) + F_y(578,5)$

$F_3 = F_x(\cos 120°) + F_y(\text{sen}\ 120°)$

$\rightarrow F_3 = F_x(-150) + F_y(259,8)$

$F_4 = F_x(\cos 230°) + F_y(\text{sen}\ 230°)$

$\rightarrow F_4 = F_x(-514,2) + F_y(-612,8)$

$R = F_1 + F_2 + F_3 + F_4$

$R_x = 1200 + 689,4 - 150 - 514,2 = 1225,2\ N$

$R_y = 0 + 578,5 + 259,8 - 612,8 = 225,5\ N$

$\boxed{R = 1245,4\ N \text{ e a direção } \theta = 10,4°}$